

Q1. DIRECTIONS *for questions 1 to 3:* Type in your answer in the input box provided below the question.

Find the sum of the digits of the least natural number N , such that the sum of the cubes of the four smallest distinct divisors of N equals $2N$.

Q2. DIRECTIONS *for questions 1 to 3:* Type in your answer in the input box provided below the question.

A cuboid whose base is a square of side 4 inches and height 6 inches is placed in a cylinder whose radius is $2\sqrt{2}$ inches and height is 6 inches. The gap between the cylinder and cuboid is filled with n spheres of radius 0.4 inches. What could be the maximum value of n ?

Q3. DIRECTIONS *for questions 1 to 3:* Type in your answer in the input box provided below the question.

If $x + y = z + 6$, where $x > 0$, $y > 0$ and $z < 0$, what is the maximum value of xy^3z^2 ?

Q4. DIRECTIONS *for questions 4 and 5:* Select the correct alternative from the given choices.

If x is a real number, then find the least possible value of the standard deviation of 23.5, 27.5, 26.5, 25.5, 24.5 and x .

- ☐ a) $\sqrt{2}$
- ☐ b) $\sqrt{\frac{20}{3}}$
- ☐ c) $\sqrt{\frac{5}{3}}$
- ☐ d) $\sqrt{\frac{14}{3}}$

Q5. DIRECTIONS for questions 4 and 5: Select the correct alternative from the given choices.

Find the value of $3 + \frac{1}{3 + \frac{3}{3 + \frac{3}{3 + \frac{2}{3 + \frac{2}{3}}}}}$.

☐ a) $\frac{2001}{613}$

☐ b) $\frac{1002}{307}$

☐ c) $\frac{1857}{569}$

☒ d) $\frac{1968}{603}$

✓ Your answer is correct

Q6. DIRECTIONS *for question 6:* Type in your answer in the input box provided below the question.

In a 1000 m long running race, if Tarun gives a head start of 40 m to Amar and still beats Amar by 10 m, then find the distance (in m) by which Tarun will beat Amar, when Amar gives a head start of 40 m to Tarun.

Q7. DIRECTIONS *for questions 7 to 9:* Select the correct alternative from the given choices.

A television company manufactures two models of televisions – A and B. Each unit of model A requires four hours to manufacture, and each unit of model B requires two hours to manufacture. The total time available in a month to manufacture these two models is 1600 hours. The profits generated on selling each unit of model A and each unit of model B are ₹1,200 and ₹1,000 respectively. Find the number of units of each of model A and model B televisions to be manufactured to maximize the profit.

- ☐ a) 200 model As and 600 model Bs
- ☐ b) Only 800 model As
- ☐ c) Only 800 model Bs
- ☐ d) None of the above

Q8. DIRECTIONS *for questions 7 to 9:* Select the correct alternative from the given choices.

In the equation $(x - m)(x - n) - 4 = 0$, if m is greater than n , which of the following is true about the roots of the equation?

- ☐ a) **Both the roots lie in $[n, m]$.**
- ☐ b) **One root is in $(-\infty, n)$ and the other root is in (m, ∞) .**
- ☐ c) **Both the roots are non-real.**
- ☐ d) **Both the roots lie in $(n + 4, m + 4)$.**

Q9. DIRECTIONS *for questions 7 to 9:* Select the correct alternative from the given choices.

The average age (in years) of a group of people is twice the number of people in the group. A person, X, leaves the group and the average age is still twice the number of people in the group. Now another person, Y, leaves the group and the average age is still twice the number of people in the group. If the ratio of the ages of X and Y is 19 : 17, then find the average age of the group, if one more person, Z, of age 16 years, leaves the group.

☐ a) 10

☐ b) 15

☐ c) 16

☐ d) 18

Q10. DIRECTIONS *for question 10:* Type in your answer in the input box provided below the question.

Four positive integers, a, b, c, d are such that a, b, c are in continued proportion and b, c, d are also in continued proportion. If $b : c = 2 : 3$, what is the minimum possible value of $(a + d)$?

Q11. DIRECTIONS *for question 11:* Select the correct alternative from the given choices.

If $n = 5$ and the angles subtended by the sectors are 70° , 71° , 72° , 73° and 74° , in how many ways can the disc be painted?

☐ a) 60

☐ b) 30

☐ c) 45

☐ d) 40

Q12. DIRECTIONS *for questions 12:* Type in your answer in the input box provided below the question.

If $n = 6$ and the angle of each sector is 60° , find the number of ways in which the disc can be painted.

Q13. DIRECTIONS *for questions 13 and 14:* Select the correct alternative from the given choices.

A mathematics teacher wrote n positive numbers on the blackboard and asked the students in her class to find the product of all the n numbers. One of her students, Rajith, randomly chose one of the n numbers and increased it by a certain quantity and found that the original product increased by 9%. Two other students, Satish and Tarun, also carried out a similar calculation and found that the original product increased by 12% and 18% respectively. If it is known that $n > 7$, and each of Rajith, Satish and Tarun increased the respective numbers that they chose by the same quantity, find the ratio of the respective numbers that they chose.

☐ a) **3 : 4 : 6**

☐ b) **6 : 4 : 3**

☐ c) **4 : 3 : 2**

☐ d) **2 : 3 : 4**

Q14. DIRECTIONS *for questions 13 and 14:* Select the correct alternative from the given choices.

Find the sum of the series

$$1 + \frac{2}{7} + \frac{4}{7^2} + \frac{7}{7^3} + \frac{11}{7^4} + \dots$$

☐ a) $\frac{301}{216}$

☐ b) $\frac{412}{343}$

☐ c) $\frac{376}{441}$

☐ d) **None of the above**

Q15. DIRECTIONS *for question 15:* Type in your answer in the input box provided below the question.

In a bag, there are a total of 150 coins in three denominations – ₹1, ₹2 and ₹5 – with at least one coin of each denomination being present in the bag. The total value of the Re.1 coins is at least 50% of the total value of the coins in the bag. If there are 23 ₹5 coins in the bag and the total value of the ₹2 coins is at least 3% of the total value of the coins in the bag, find the number of ₹2 coins in the bag.

Q16. *Directions for question 16:* Select the correct the alternative from the given choices.

Ramu and Somu start running simultaneously from two ends of a track in opposite directions and with speeds in the ratio of 2 : 3 respectively.

Every time they meet, they interchange their speeds and also reverse their directions. At how many distinct points on the track do they meet each other, if they run continuously between the two ends of the tracks?

☐ a) 3

☐ b) 4

☐ c) 5

☐ d) 6

Q17. DIRECTIONS *for question 17:* Type in your answer in the input box provided below the question.

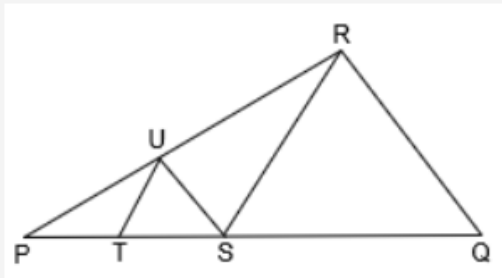
Six men and fourteen women can complete a work in five days, whereas two men and three women can complete one-fourth of the same work in four days. If one man and two women take up and complete the same work, earning a total wage of ₹11,791 for the same, what is the total share (in ₹) of the two women in this amount?

Q18. DIRECTIONS *for questions 18 and 19:* Select the correct alternative from the given choices.

There are M positive numbers and the average of each possible pair of these numbers is found. If the average of all these averages is N , what is the average of the M numbers?

- ☐ a) $\frac{N}{M}$
- ☐ b) $\frac{M+N}{2}$
- ☐ c) $M + N$
- ☐ d) N

Q19. DIRECTIONS for questions 18 and 19: Select the correct alternative from the given choices.



In triangle PQR, T and S are points on PQ and U is a point in PR such that UT and RS are parallel and US and RQ are parallel. If $PS : SQ = 2 : 3$, find $TS : SQ$.

- ☐ a) 3 : 5
- ☐ b) 2 : 5
- ☐ c) 4 : 5
- ☐ d) 2 : 3

Q20. *Directions for question 20:* Type in your answer in the input box provided below the question.

Find the maximum possible remainder if $(104)^n$ is divided by 51, given that n is a natural number greater than 10.

Q21. DIRECTIONS *for questions 21 and 22:* Select the correct alternative from the given choices.

The cost (in ₹) of a watermelon is proportional to the square-root of its weight (in pounds). If a watermelon weighing 16 pounds costs ₹50, how much will a watermelon costing ₹37.5 weigh, in pounds?

- ☐ a) 4
- ☐ b) 8
- ☒ c) 9 ✓ **Your answer is correct**
- ☐ d) 12

Q22. DIRECTIONS *for questions 21 and 22:* Select the correct alternative from the given choices.

An equilateral triangle ABC is drawn on a sheet of paper. A circle is then drawn with BG as a diameter, where G is the centroid of the triangle ABC. The circle is then cut away. The part of the triangle which is on the circle thus removed, is now cut away. If the side of the triangle is $2\sqrt{3}$ units, what is the approximate area of the circle (in sq. units) that remains?

- ☐ a) **0.96**
- ☐ b) **1.23**
- ☐ c) **1.57**
- ☐ d) **1.96**